

From experimental autonomy to scientific understanding in programmable chemical worlds

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Abstract

Self-driving laboratories increasingly choose which experiments to perform and which evidence to pursue. Yet equipment and workflow constraints, experimental costs, safety requirements and unknown mechanisms make their scientific learning difficult to examine systematically in physical laboratories. We introduce ChemWorld, which combines experimental freedom with independent evaluator control in composable, programmable chemical worlds: agents assemble operations on evolving samples, while evaluators separately configure process laws and supplied information. Cross-system studies and targeted comparisons of five model configurations connect autonomous research histories to operating outcomes and sealed predictions under new interventions. Better experimental outcomes need not accompany more accurate predictions. In one equilibrium configuration, correct partial priors improve prediction in one region but increase error 8.7-fold under extreme dilution relative to withholding those priors. Across the five configurations, the same task elicits distinct patterns of evidence acquisition and extrapolation. Accurate forecasts occur both without direct sampling of the new regime and after autonomous experiments reveal its changed response. Crystallization exposes a complementary departure: predictions systematically underestimate purity even though experiments establish a nearly constant level. These findings reveal different ways autonomous researchers extend and depart from experimental evidence. They establish programmable chemical worlds as an instrument for studying scientific learning and motivate experimental planning that tests when acquired relationships should persist or be revised.

Introduction

Self-driving laboratories extend experimental science through automated execution and adaptive experimental planning^{1,2}. Robot scientists have generated and experimentally tested biological hypotheses³, while subsequent platforms have demonstrated autonomous materials search and reaction optimization⁴⁻⁷. Mobile robots also support exploratory synthesis using complementary analytical instruments⁸. Language-model agents connect scientific instructions, tools and laboratory operations^{9,10}, and multi-agent research teams have designed biomolecules subsequently validated in experiments¹¹. As systems gain the freedom to choose operation sequences, intermediate measurements and subsequent experiments, they increasingly shape their own research paths. The scientific opportunity extends from executing procedures and optimizing conditions to deciding which evidence to acquire and what that evidence implies.

Greater experimental freedom also makes scientific learning harder to evaluate. Different research paths can support the same operating success while exposing different aspects of a chemical system. A productive recipe may follow from a transferable explanation or from a locally effective empirical rule. Decision-focused learning likewise distinguishes prediction error from the quality of decisions made using

predictions¹². Physical laboratories offer rich possibilities for discovery, while each implemented workflow has practical limits imposed by equipment, materials and procedures. Unknown mechanisms, evolving material states and the cost of repeated experiments complicate controlled comparisons of how autonomous researchers acquire and use evidence. Understanding their progress therefore requires connecting experimental achievements to the histories and judgments that produced them.

We study scientific learning by connecting knowledge acquired through autonomous experiments to predictions under new interventions. The same researcher’s experimental history and public scientific account reveal which evidence was acquired and how it was extended to unmeasured conditions. Information-gain criteria and adaptive design policies formalize the selection of informative experiments^{13–15}, while strong inference emphasizes experiments that discriminate competing hypotheses¹⁶. These traditions motivate evaluating evidence acquisition together with the conclusions drawn from it.

Existing benchmarks evaluate complementary parts of this process. ChemBench assesses chemical knowledge and reasoning through question answering¹⁷, and DiscoveryBench evaluates hypothesis discovery from supplied datasets¹⁸. Optimization benchmarks offer repeatable objectives^{19,20}. Interactive chemistry and discovery environments support multi-step experimentation^{21–23}; BoxingGym connects experimental design to model discovery and explanation-based prediction²⁴. Science-Gym evaluates active data collection and equation discovery²⁵, while PhysGym explicitly varies prior knowledge in interactive physics discovery²⁶.

ChemWorld brings this investigation into composable chemical processes with persistent samples and flexible operations. Evaluator control over compatible world laws and supplied information supports matched comparisons of research conditions. We follow the same researcher from self-chosen experimental operations to independently retested recommendations and sealed predictions under new interventions. Complete operational histories connect these outcomes to the evidence the researcher actually acquired.

Our results connect two features of autonomous scientific learning. Experimental success and predictive accuracy can rank the same researchers differently. Moreover, researchers facing the same chemical worlds acquire different evidence and follow distinct routes to extrapolation: some extend a local plateau into conditions where it fails, others extrapolate accurately from local observations, and others experimentally encounter a new response before predicting it. Crystallization reveals the complementary difficulty of preserving an experimentally established relationship when it remains stable. Following these paths makes it possible to study how autonomous researchers decide which relationships to retain and which to revise.

Results

Tracing autonomous experimentation in controlled chemical worlds

ChemWorld constructs chemical processes as composable, executable experimental worlds. Components for reaction, thermal processing, phase equilibrium, separation and crystallization connect through compatible interfaces to maintain material, energy and sample states. An environment can support characterization of a single process or connect reaction with extraction, washing, concentration or crystallization in a multistage workflow, allowing upstream operations to affect downstream processing and final measurements. Configuring process components, their connections, model parameters and

available instruments creates different research environments; adding process models and observation components extends the platform.

Agents organize their own experiments in these worlds: they select materials and loadings, adjust operating parameters and procedure order, and use intermediate results to decide whether to continue processing, repeat an experiment or terminate it. Sample states persist between operations, so reaction, thermal and separation histories jointly determine subsequent outcomes. Measurements must also be actively selected. Observation channels differ in information, noise and resource cost, requiring researchers to consider both experimental progress and evidence acquisition. Evaluators independently configure compatible process laws and public information, supporting comparisons of research conditions within the same world or changes to hidden laws under a shared public interface. Complete operation, observation and resource records connect choices to consequences and support exact replay of environment trajectories (Fig. 1).

The cross-system study uses GPT-5.6 Sol with medium reasoning effort for 240 autonomous campaigns across six chemical system families. The equilibrium parameter task includes a further 60 campaigns from four model configurations to compare evidence acquisition and extrapolation. Each study block contains five worlds and three information conditions: Opaque supplies no additional instance-specific prior, Aligned supplies correct but incomplete information, and MisIndexed supplies a fixed mismatch; all arms retain the public operational contract needed to conduct experiments. Where required, sealed recommendations receive an independent retest. The original researchers retain their experimental histories and subsequently provide scientific accounts, sealed predictions under new conditions and retrospective reflections, without reference-outcome feedback. Operating outcomes and predictive judgments can thus be traced to evidence acquired by the same researcher. The complete matrix, resources and assessment protocols are specified in Methods.

A twelve-batch crystallization campaign illustrates how a researcher adjusts an experimental procedure (Fig. 2). The first four batches fail because of excessive fines. Batch 5 changes several conditions, including cooling and seeding order, and first obtains a feasible result. Recovery reaches 45.5% at batch 8; the sealed recipe achieves 43.9% on independent retest while continuing to meet the quality constraints. A subsequent solvent reduction again violates those constraints. By choosing successive interventions, the researcher shapes both its experimental outcomes and the evidence available for subsequent scientific judgments.

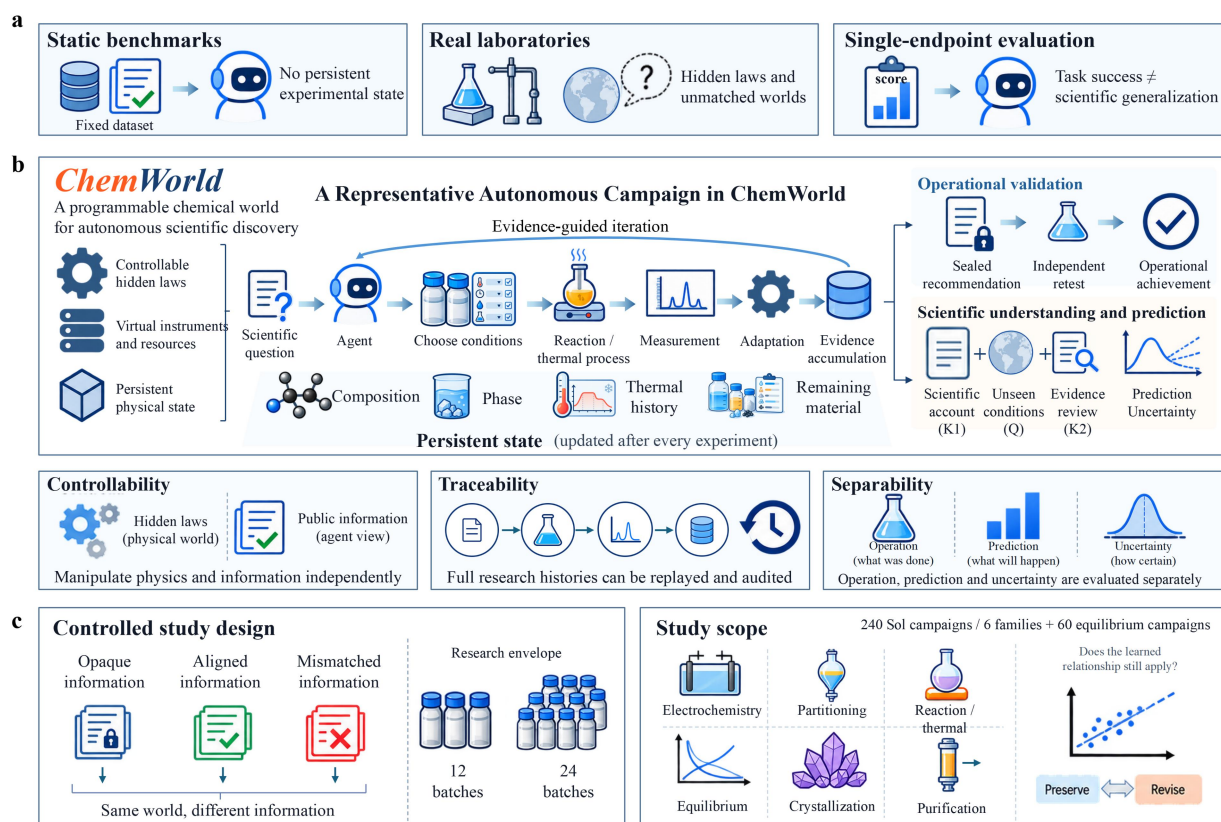


Figure 1. ChemWorld as a controlled experimental instrument for autonomous science. **a**, Evaluation challenges arising from fixed datasets, physical laboratories and single-endpoint scoring. **b**, Autonomous research connects experiment selection, process execution, measurement and evidence-informed adaptation within a stateful physical world. Hidden laws and public information can be varied separately; research histories can be replayed; operational performance, prediction and uncertainty are evaluated separately. Where required, operating recommendations are sealed and independently retested. The original session then retains its research history while providing a scientific account (K1), sealed predictions (Q) and retrospective evidence review (K2). Neither Q nor K2 receives target-outcome feedback. **c**, Information conditions in matched worlds, independent 12- and 24-batch resource envelopes, and 240 Sol-medium campaigns across six families plus 60 targeted equilibrium campaigns using four additional model configurations. “Mismatched information” denotes MisIndexed. Budget comparisons apply only to systems with both resource envelopes. Methods and Supplementary Tables A1 and F3 report the study matrix.

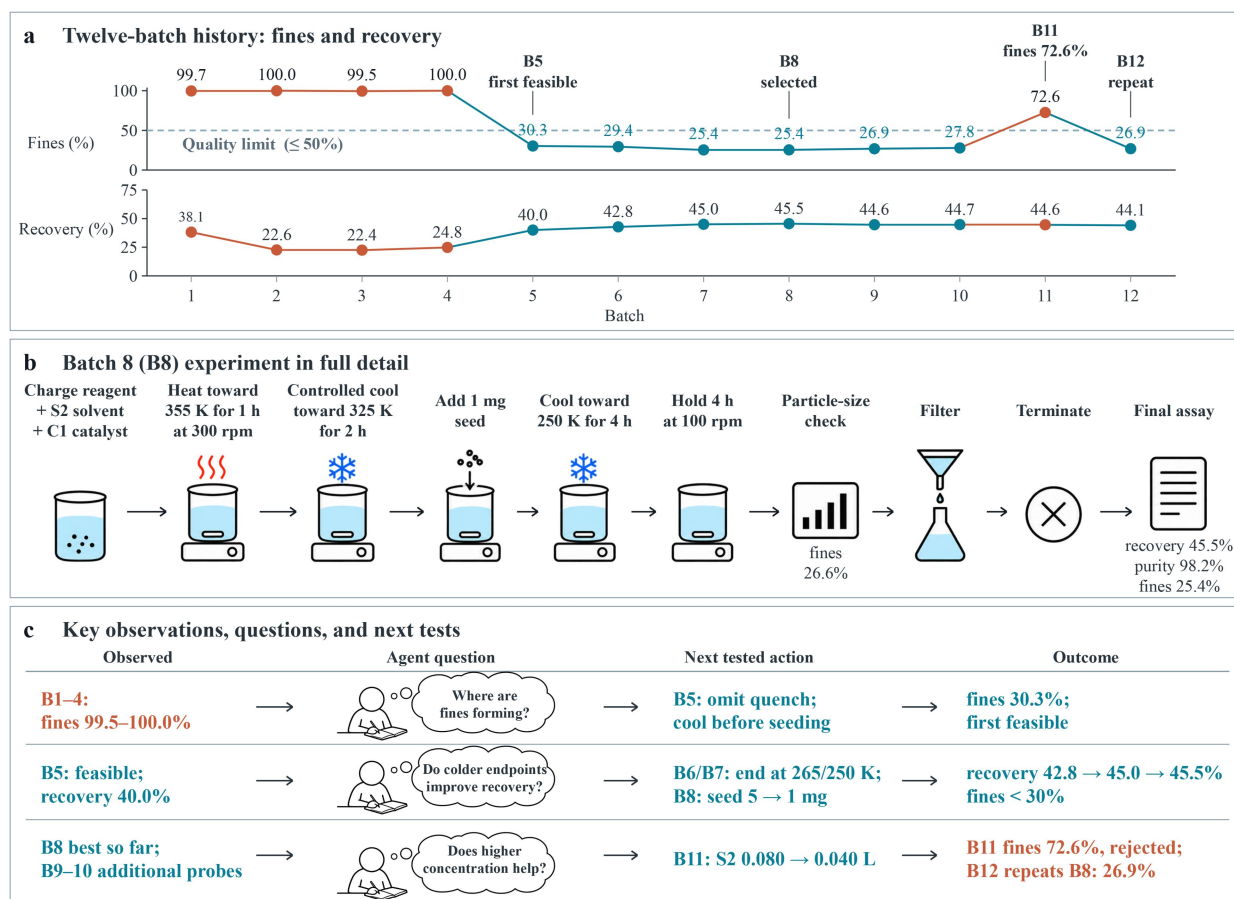


Figure 2. One complete autonomous crystallization campaign. a, Final-assay outcomes for all twelve source batches in the W04/MisIndexed session. Teal indicates quality-conforming results and ochre indicates failures; the dashed line marks the 50% fines limit. Batch 5 is first feasible, batch 8 is sealed as the recommendation, batch 11 violates the fines constraint and batch 12 repeats the selected recipe. b, All twelve operation calls in batch 8, grouped into ten visible stages. The intermediate particle-size check reports 26.6% fines and the final assay 25.4%; temperatures are requested targets. Independent retesting uses the sealed recipe with a different observation seed. c, Three observation–subsequent action–outcome sequences. Questions are reconstructed by the authors from operation records; K1 was written after experimentation. Recovery excludes seed contributions. Feasibility requires purity of at least 80%, fines of at most 50% and particles to be present; fines are particles below 20 micrometres. Supplementary Methods describe case selection and interpretation of the process changes.

Operational and predictive gains need not coincide

We first ask whether researchers that deliver better experimental procedures also predict new conditions more accurately. In electrochemical and reaction tasks that allow material selection and adjustment of electrical or thermal conditions, researchers prioritize either developing and testing scientific understanding (discovery) or obtaining high-scoring procedures (optimization). Independent campaigns under matched conditions are evaluated through recommendation retests and score predictions under new conditions (Fig. 3). Optimization achieves a higher retest score in 26 of 30 electrochemical pairs: prediction mean absolute error (MAE) is also lower in 13, but higher in the other 13. In reaction processing, discovery obtains both better retests and better predictions in 17 of 30 pairs. The two endpoints distinguish achievements that a single measure of experimental success would conflate.

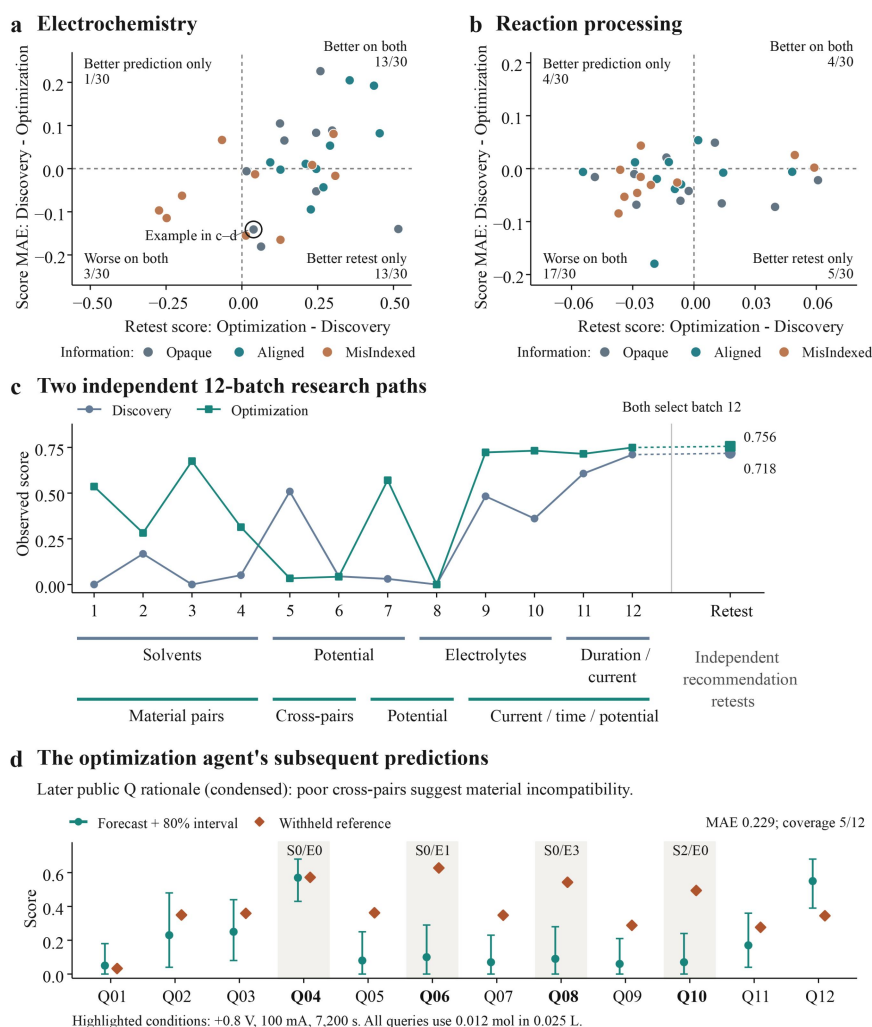


Figure 3. Research objectives, operational outcomes and predictive judgments. a,b, Thirty matched campaign pairs each in electrochemistry (a) and reaction processing (b), spanning five worlds. Positive differences favour optimization on the respective endpoint; quadrant labels identify the better commission and the ring marks the selected case. c, All batch scores in two independent twelve-batch campaigns, recorded operation changes and independent retests of sealed recommendations. Lines join successive observations; discovery batches 11–12 include segmented electrolysis and intermediate measurement. d, All twelve original optimization forecasts, nominal 80% prediction intervals and withheld reference observations. The rationale condenses the public account supplied with the subsequent Q forecasts. Shading identifies four queries at common electrical settings with different material pairs, retaining an accurate forecast alongside underestimates. Pairing fixes world, information arm, budget and prior locus. Methods define the counts and reference targets.

One electrochemical pair connects this contrast to actual research paths (Fig. 3c,d). Discovery first compares materials and conditions, then increases duration and the current cap; optimization screens material combinations before adjusting operating conditions for a promising pair. Optimization ultimately delivers the higher-scoring retested procedure, 0.756 versus 0.718 for discovery, but also has higher prediction MAE, 0.229 versus 0.088. After several cross-pair experiments perform poorly, the optimization researcher's public prediction rationale classifies these combinations as low-activity. For one previously tested combination under new conditions, it predicts 0.070 against a reference of 0.494; for another combination, prediction and reference are 0.570 and 0.572. At the end of two independent campaigns, operating and predictive outcomes therefore give opposite rankings: optimization's recommendation has the higher retest score, while discovery predicts new conditions more accurately.

Increasing research resources likewise does not synchronize the two benefits. Independent 12- and 24-batch campaigns show improvements in some electrochemical, partitioning and crystallization predictions, but the number of feasible crystallization recommendations does not increase, nor does mean retested recovery (Supplementary Fig. S4). Experimental resources can improve prediction without simultaneously improving the final procedure delivered.

Relevant scientific knowledge can also provide substantial help. In the reaction parameter-prior study, Aligned reduces macro prediction MAE from 0.102 to 0.057 for discovery and from 0.139 to 0.084 for optimization, outperforming Opaque in four of five worlds under each commission (Supplementary Fig. S2a,b). Resources and knowledge therefore offer concrete routes to predictive improvement, alongside the procedural achievements revealed by retesting. The next question is how researchers carry experimentally acquired relationships into conditions beyond their observations. The equilibrium task makes this transition directly visible.

The same chemical task elicits distinct routes to extrapolation

Researchers can vary loading, dilution and addition order in the equilibrium task, then predict pH, dissociation fraction and precipitation response under new conditions. The process laws remain fixed, but further dilution changes the relative contributions to charge balance, taking dissociation away from its local high-concentration plateau. Across Sol’s fifteen campaigns, all 180 source assays have nominal concentrations above 0.018 M and dissociation fractions of 5.5–9.2%; reference means at the lowest-concentration test condition reach 44.8–71.6% across the five worlds (Fig. 4a). This separates relationships the researcher actually observed from the extrapolation required by new conditions.

In the Sol campaigns, correct prior information reverses its predictive benefit across test regions. Aligned reduces macro MAE on the remaining nine queries from Opaque’s 0.01008 to 0.00465, but raises error on the three lowest-concentration queries from 0.01833 to 0.15862, approximately 8.7-fold; both directions occur in all five worlds (Fig. 4b; Supplementary Table F2). Nominal 80% interval coverage on the three lowest-concentration queries also falls from 93.3% for Opaque to 2.2% for Aligned. Original answers show Opaque extrapolating an increase in dissociation without directly sampling dilute solutions, while Aligned mainly extends the plateau or a weak local trend (Supplementary Fig. S8c). One Aligned answer acknowledges that the plateau need not persist towards zero loading, yet predicts 8.04% [5.30%, 10.80%] against a reference mean of 65.46%. The explanation acknowledges a boundary to applicability, but the sealed numerical forecast continues to use the plateau level.

The five model configurations exhibit distinct patterns of generalization under the same scientific conditions (Fig. 4c; Supplementary Table F3). We compare Sol with the medium configurations of GPT-5.5, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra in the same five worlds, three information conditions and twelve-batch envelope. Whereas Sol shows a pronounced reversal of prior benefit, Luna and Terra obtain lower mean error with Aligned in both query groups, and GPT-5.5 changes little. Astra maintains low error across all three information arms on the lowest-concentration queries, with MAE of 0.01857–0.04040 and original interval coverage of 82.7–92.4%. A shared task and information protocol thus elicit different responses to scientific knowledge, from gains across both query groups to a reversal between them.

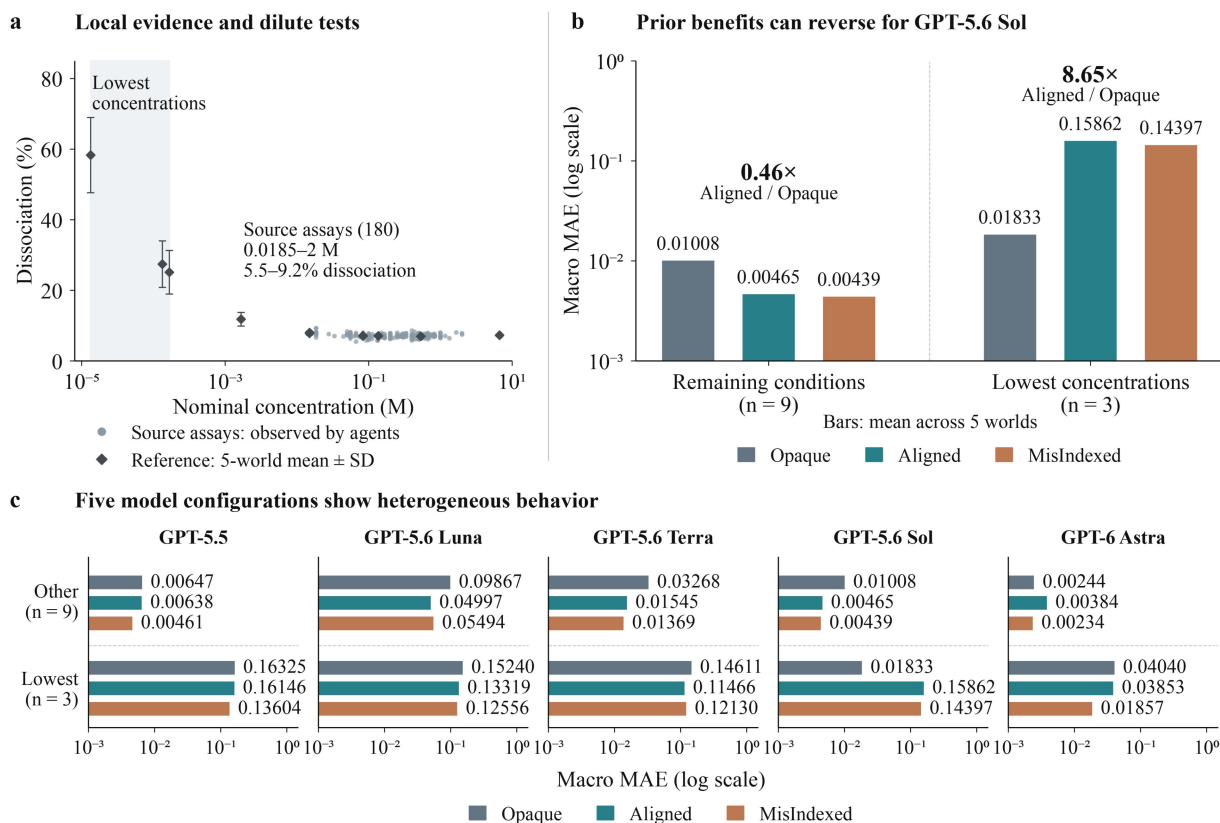


Figure 4. Local evidence, reversal of prior benefit and distinct extrapolation patterns across models. a, Sol’s 180 source assays and twelve withheld conditions, with reference means and sample standard deviations across five worlds. Shading marks the three lowest-concentration conditions. b, Sol macro MAE for the remaining nine and lowest three conditions; ratios compare Aligned with Opaque. c, Five model configurations under the same world, information and budget protocol, with three information arms in each query group. Bars in b,c show five-world means on logarithmic axes starting at 0.001; macro MAE equally weights normalized pH, dissociation fraction and precipitation signal. Statistical units, subgroup construction and complete values are provided in Methods and Supplementary Tables F2–F3.

The experimental histories reveal a corresponding difference in evidence acquisition. Eleven of Astra’s fifteen campaigns assay positively loaded samples at nominal concentrations no greater than 1 mM, compared with zero of fifteen for each of the other four configurations (Fig. 5a). Sol/Opaque nevertheless predicts the dilute response accurately from its local observations, achieving MAE of 0.01833 on the three lowest-concentration queries (Fig. 5b,c). Accurate extrapolation therefore occurs even in campaigns that never directly sample this regime.

An Astra campaign shows the complementary route of acquiring new evidence before extrapolation (Fig. 5d). In World 3/MisIndexed, dissociation is 6.3–7.4% in the first three batches. Batch 5 lowers concentration to 18.5 μ M and observes 53.8%; later batches return to higher concentrations and explore staged dilution, additional loading and longer waiting. The subsequent public scientific account distinguishes the high-concentration plateau from the low-concentration response. The researcher then seals a forecast of 58.80% [54.20%, 63.40%] at 13.3 μ M, against a reference mean of 57.97%. The recorded sequence connects autonomous observation of a changed response to its recognition in the scientific account and an accurate prediction at a new condition.

Prediction accuracy alone obscures these different research paths. Sol/Opaque extrapolates accurately beyond its observations, while the Astra case first acquires evidence outside the plateau and subsequently

predicts the changed response. Linking forecasts to experimental histories reveals these distinct routes to a successful outcome alongside the local-rule extension that fails under dilution.

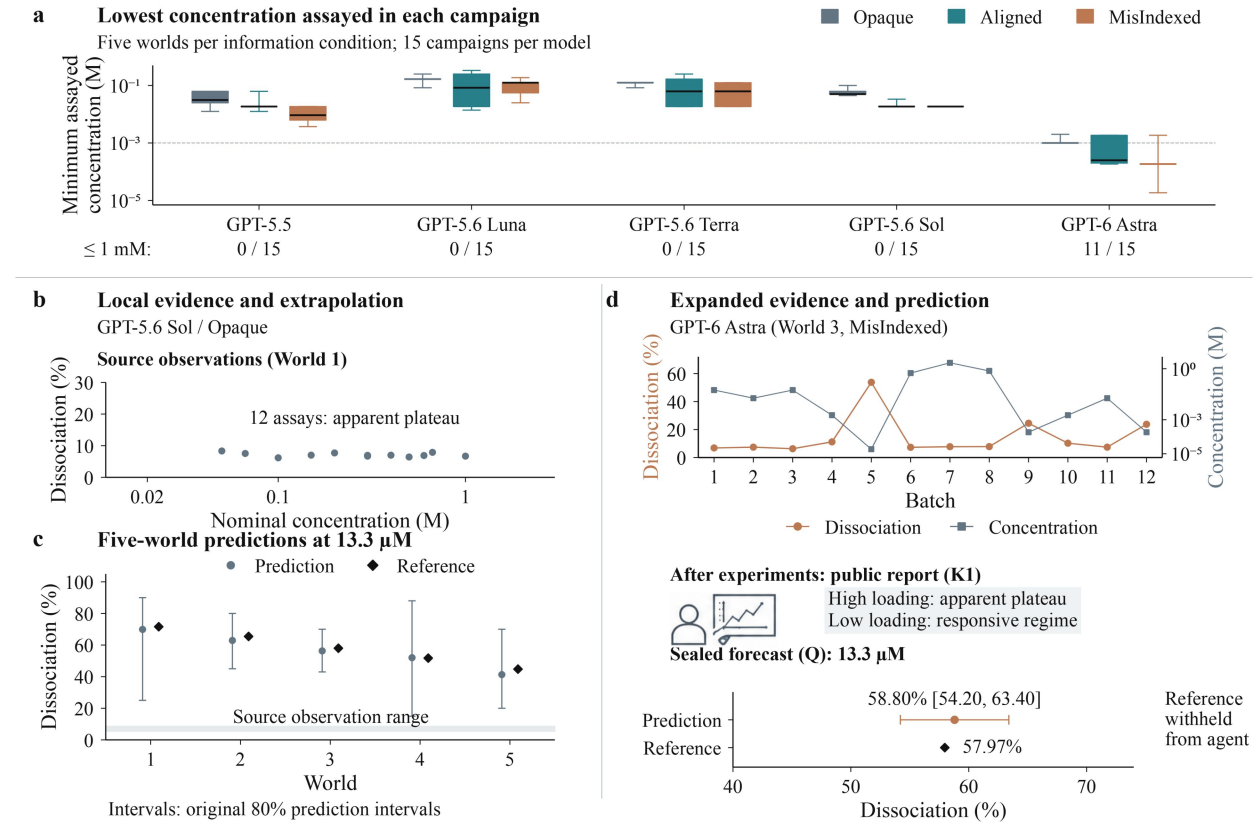


Figure 5. Different routes from autonomous experiments to accurate extrapolation. a, Lowest positively loaded nominal concentration assayed in each campaign. Each model has three information-arm boxplots, with five worlds per box: original-scale quartiles, median and minimum–maximum whiskers, displayed on a logarithmic axis. The dashed line and counts mark the descriptive 1 mM threshold. b, Twelve Sol World 1/Opaque assays show the locally observed plateau. c, Sol/Opaque forecasts at 13.3 μM across five worlds, with original 80% prediction intervals and reference means; shading spans dissociation observations in these source campaigns. d, All twelve batches of the Astra World 3/MisIndexed campaign, showing dissociation (left axis) and nominal concentration (right, logarithmic axis), followed by its post-experiment K1 account and sealed Q forecast at 13.3 μM. The displayed account condenses the public report; the forecast retains its original 80% interval. These retrospectively selected cases connect acquired evidence to subsequent predictions; Methods describe case selection and trajectory analysis.

Predictions systematically underestimate stable purity

The equilibrium studies expose the challenge of anticipating a response that changes beyond the observed region. Sol’s crystallization campaigns reveal the complementary difficulty of preserving a relationship that remains stable. Mean purity is 98.53% in source observations and 98.51% in withheld references; the mean within-world range across withheld conditions is only 2.44 percentage points. Yet mean predictions are 93.54% and 94.89% at 12 and 24 batches, respectively, with 335 of 360 predictions below their reference values (Fig. 6a).

Within the same campaigns, the recovery–purity comparison shows that this departure is response-specific. A baseline repeats each campaign’s own public final-assay mean for every withheld condition, using no additional experiments or withheld outcomes. Agent recovery predictions outperform this baseline in 26 of 30 campaigns, whereas purity predictions outperform it in none (Fig. 6b); mean purity MAEs are 4.35 and 0.48 percentage points, respectively. The same researcher can predict changing recovery more effectively than the mean baseline, yet fail to preserve the stable purity level already established by its experiments.

Purity errors predominantly reflect an overall downward shift. Mean signed error is -4.29 percentage points, and campaign-level offsets account for 87.6% of total squared error (Supplementary Table G2). Information conditions and research budgets alter the relative performance of recovery and purity forecasts, while prediction intervals fail to accommodate the purity errors reliably (Fig. 6c,d). Aligned and MisIndexed report wider mean intervals than Opaque yet achieve poorer coverage. The main error is thus a departure from the empirically established purity level, accompanied by errors in predicting its smaller variations across conditions.

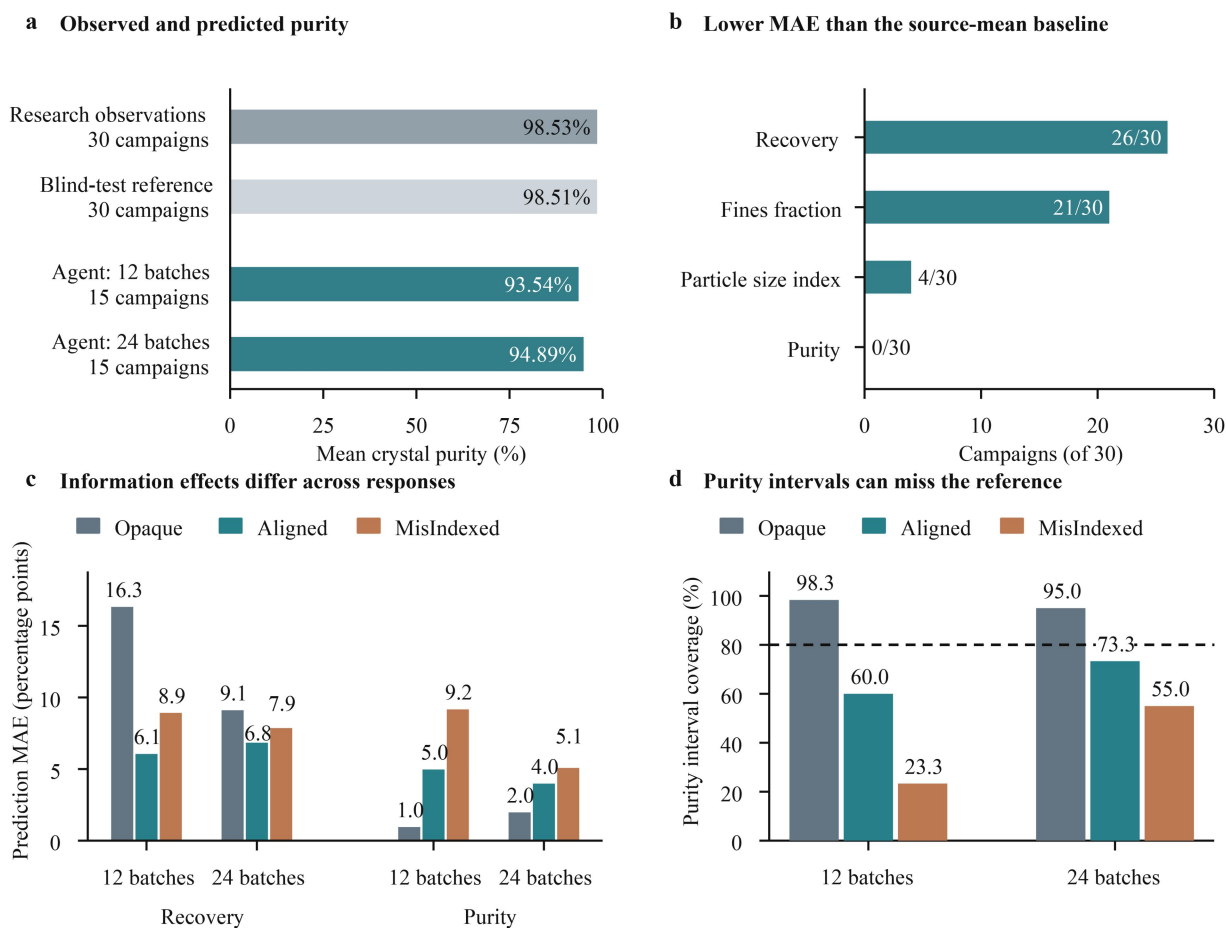


Figure 6. Crystallization forecasts depart from experimental evidence in a response-specific manner. a, Mean purity in source observations and withheld references, each from thirty campaigns, compared with forecasts at two budgets, each with fifteen campaigns. Bars equally weight campaign means. b, Campaigns with lower agent MAE than the source-mean predictor, compared separately within each response. The baseline uses the same campaign’s public final-assay mean for every withheld recipe. c, Recovery and purity MAE by information arm and budget, with five worlds per bar. d, Coverage of original nominal 80% purity intervals by information arm and budget, aggregated over five campaigns with twelve queries each; the dashed line marks the nominal level. Methods define statistical units, baselines and inclusion.

Discussion

Experimentally acquired relationships can be extended too far or insufficiently preserved. In Sol/Aligned equilibrium campaigns, predictions extend a local plateau into dilute conditions where the response changes substantially. In crystallization, predictions fall systematically below the stable purity jointly established by experiments and withheld outcomes. These opposite departures make scientific generalization concrete: which experimental relationships should change with an intervention, and which should continue to hold²⁷?

The model comparison reveals distinct scientific behaviours within the same experimental setting. Sol/Opaque extrapolates successfully from local observations, while Astra frequently acquires evidence at lower concentrations and, in the selected campaign, observes the changed response before accurately predicting it. Similar predictive success can therefore accompany different evidence histories. Conversely, one Sol/Aligned answer describes a boundary to the plateau in words but retains its level numerically: articulating an applicability limit and applying it in a quantitative judgment are distinct achievements. ChemWorld makes these differences visible by connecting experimental choices, acquired evidence, public accounts and sealed forecasts. Evaluating scientific learning requires this connection between research process and outcome, alongside tests of uncertainty under changing conditions^{28,29}.

These observations suggest a design principle for scientific agents: make the conditions under which a learned relationship persists or changes an explicit target of experiment selection. Experiments can then be chosen to distinguish competing explanations¹⁶. The Astra trajectory provides an executed example of observing a response beyond the local plateau before sealing predictions. Some Sol researchers propose trace-loading experiments only during retrospective review, and a separate crystallization pair proposes a thermal test at the same later stage (Supplementary section C.4 and Fig. S5); these proposals remain unexecuted. The design opportunity is to bring these judgments forward, so they guide evidence acquisition before a forecast is sealed.

Cross-system results cover one model configuration across six families; the five-model equilibrium comparison reuses worlds and readouts, with one campaign per model–world–information condition and exploratory subgroup analyses. Separating the contributions of sampling, prior use and interpretation, and testing transfer to physical laboratories, remain future work. Our measures assess predictions under new interventions and public scientific accounts; complete mechanistic understanding and internal belief revision require further evidence³⁰. Programmable worlds make autonomous research behaviour an object of controlled investigation, connecting what researchers achieve to the evidence they acquire and how they extend it to new conditions.